Proportions on one page

Hi! Bulya and I drew everything important about proportions. Follow the arrows!



Ratio 1

Divide one number by the other. We read it as "one to two".

$1 : 2 = 0.5$

1 → half of 2

2

Proportion 2

Two ratios that equal the same number. We put "=" between them.

$1 : 2 = \frac{4}{7} : \frac{8}{7}$

0.5

Outer and inner numbers 3

The **outer** numbers sit at the ends, the **inner** ones in the middle.



The outer ones are like me and Ponya at each end of the sofa, and the inner ones squeeze in between!

outer

$a : b = c : d$

inner

The big rule: cross-multiply 4

Write it as fractions and multiply across. **Outer numbers multiplied = inner numbers multiplied.** Equal? Then it's a proportion!

$\frac{a}{b} = \frac{c}{d} \rightarrow a \times d = b \times c$

$\frac{10}{40} = \frac{6}{24} \rightarrow 10 \times 24 = 240,\ 40 \times 6 = 240$ ✔

Finding the missing number 5

1. Multiply the pair that doesn't have x (both outer or both inner).
2. Divide by x's partner in its pair.

① $12 \times 5 = 60$

$12 : x = 4 : 5$

② $60 \div 4 = 15$

① $8 \times 15 = 120$

$5 : 8 = 15 : y$

② $120 \div 5 = 24$

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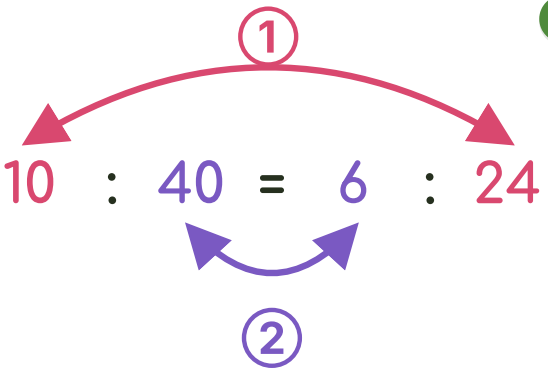
Swapping to make new proportions

Swap places and it's still a proportion, because the products stay the same.

① swap the outer numbers: 24 : 40 = 6 : 10

② swap the inner numbers: 10 : 6 = 40 : 24

①+② swap both pairs: 24 : 6 = 40 : 10

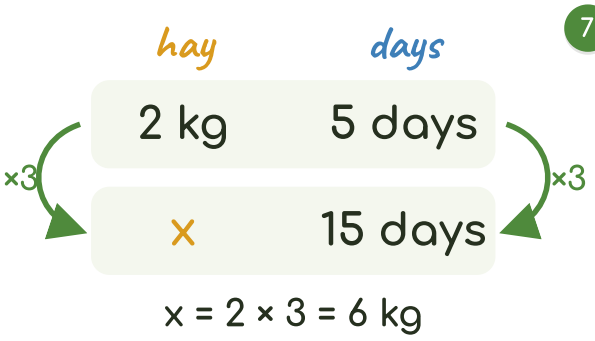


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Proportion in real life

2 kg of hay lasts 5 days. How much hay for 15 days? 3 times as many days, so 3 times as much hay.

2 : 5 = x : 15 → x = (2 × 15) ÷ 5 = 6 kg



x = 2 × 3 = 6 kg



Found the answer? Put it back in and cross-multiply. If it matches, well done!

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Built on his slides and lessons. Made by Mykola Stepanov for his daughter Varia